

Hubble Tension

William, Sophie, Daniel, Thomas, Grant

Short Explanation

The Hubble tension refers to the incompatibility of two different ways to acquire the value for the Hubble constant (H_0), a constant showing how fast things in the universe are receding with relation to their distance.

Expansion of Universe

1. Big Bang

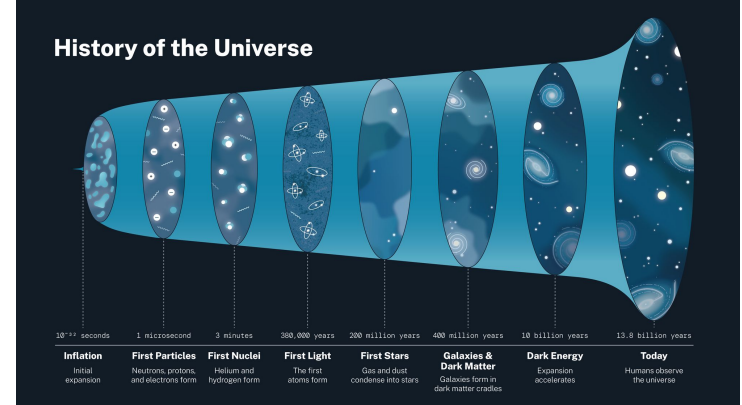
-13.8 billion years ago, all matter and energy were compressed into singularity. This point rapidly expanded, cooling over time to form space, time, and the cosmos we observe today.

2. At first, expansion decelerated due to matter

3. Then, accelerated due to dark energy

-the universe shall be decelerating expanding due to the gravity,
but actually it is accelerating

-Dark energy features negative pressure



NASA. "History of the Universe." *NASA Science*, NASA,
<https://science.nasa.gov/resource/history-of-the-universe/>.
Accessed 12 July 2026.

Definition of Hubble Constant

$a(t)$ is the scale factor of the size of the universe

Hubble constant is technically not a constant

But rather is changing around $-3.2 * 10^{-9} \% \text{ year}$

However, this result does not explain the discrepancy between the two results as the error margin does not expand to that range, and for our presentation we will only focus on the value within constant range, denoted H_0 .

Hubble Constant

$$v = H_0 d$$

v : recessional velocity [LT⁻¹]

H_0 : Hubble constant [T⁻¹]

d : distance [L]

So by knowing the recessional velocity and distance of a distant galaxy, we can calculate the Hubble constant.

Hubble Time

For now, we will just put $H_0 \approx 70 \text{ km/s/Mpc}$

If we assume constant expansion of universe, the age of the universe can be calculated as

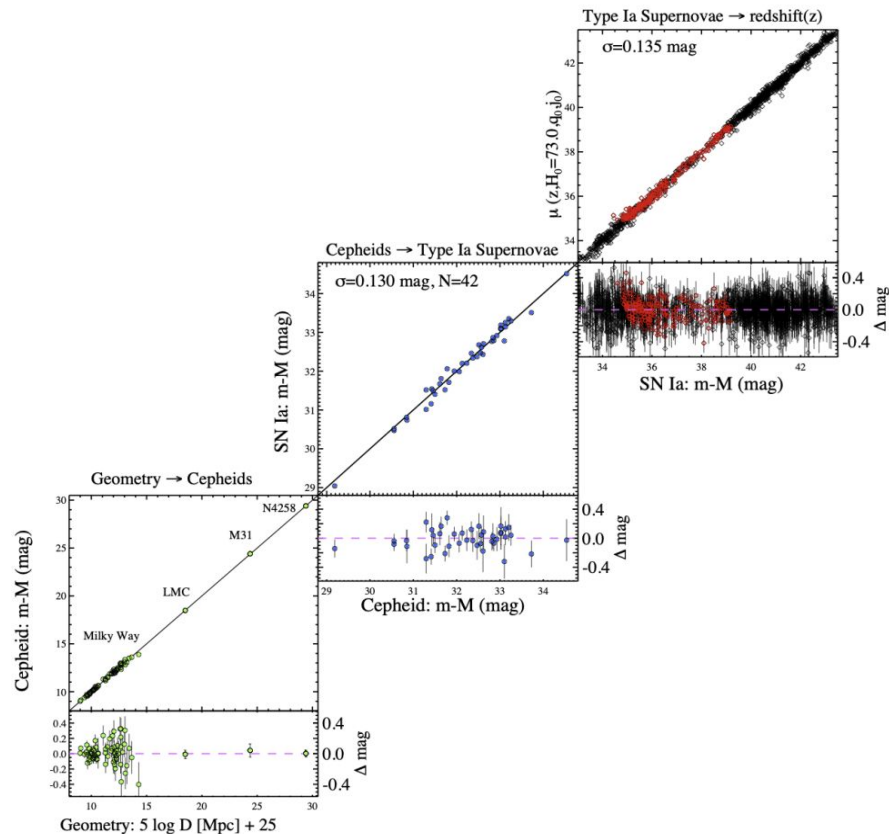
$$t = \frac{d}{v} = \frac{d}{H_0 d} = \frac{1}{H_0}$$

$$H_0 = \frac{70 \text{ km/s}}{1 \text{ Mpc}}, \quad \frac{1}{H_0} = \frac{1 \text{ Mpc}}{70 \text{ km/s}} = \frac{3.09 \times 10^{19} \text{ km}}{70 \text{ km/s}} \approx 4.4 \times 10^{17} \text{ s} \approx 14 \text{ billion years}$$

However, this is a naive estimation since the expansion actually decelerated and accelerated, approximately equaling Hubble time.

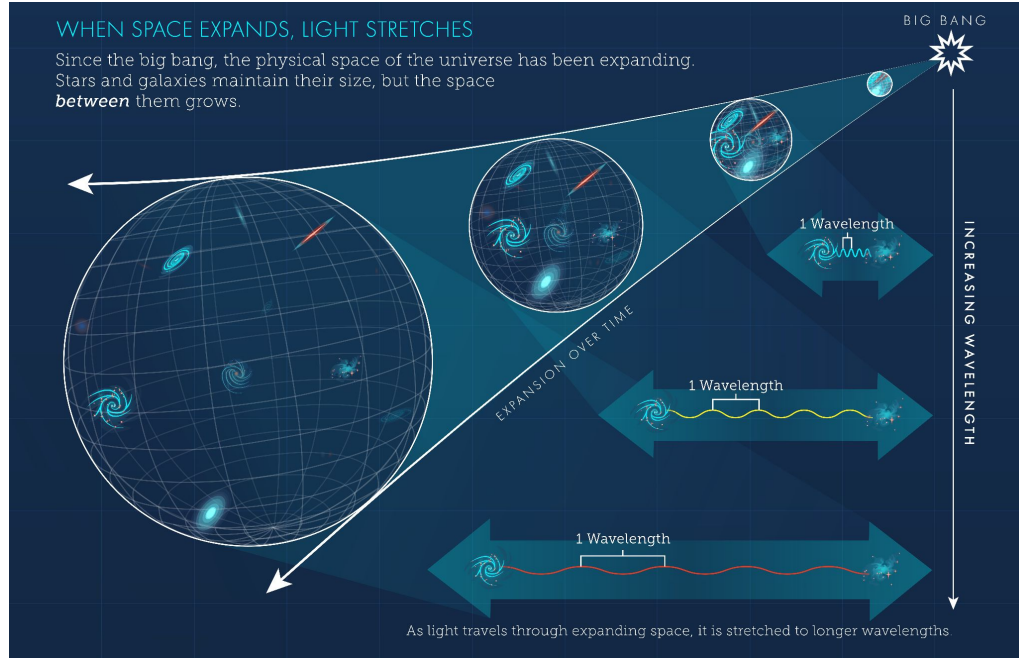
Cosmic Distance Ladder

1. Cosmological Redshift
2. Distance
 - Parallax
 - Cepheid Stars
 - Type Ia Supernova



Kamionkowski, Marc, and Adam G. Riess. "The Hubble Tension and Early Dark Energy." *Annual Review of Nuclear and Particle Science*, vol. 73, no. 1, Sept. 2023, pp. 153–80, <https://doi.org/10.1146/annurev-nucl-111422-024107>. Accessed 13 July 2026.

Cosmological Redshift



NASA. "Hubble Cosmological Redshift - NASA Science." *Nasa.Gov*, 17 Sept. 2024, <https://science.nasa.gov/mission/hubble/science/science-behind-the-discoveries/hubble-cosmological-redshift/>. Accessed 12 July 2026.

- Light (EM waves) from different galaxies appear to have a longer wavelength
- Not the same as traditional Doppler effect (changing wavelength due to objects physically moving)
- Occurs due to spacetime expansion, which stretches the wavelength longer (the red side)
- However, the observed effect is in essence the same as the Doppler effect, as the observed appears to move away.

Cosmological Redshift

We are able to calculate the amount of redshift since we know the emitted wavelengths of known elements (such as hydrogen), and compare them to a laboratory produced source with the original wavelength by matching the plasma composition of the star.

And using the redshift we can calculate the receding velocity of the cosmological object by using the difference in wavelength (λ) and the speed of light (c).

$$z = \frac{\lambda_{obs} - \lambda_{emit}}{\lambda_{emit}}$$

$$v = cz$$

For only small z (redshift), which is true for obtaining Hubble constant

Distance - Geometry (Parallax)

We use simple geometry to calculate the distance between the Earth and a celestial body that is close enough for calculation.

$$p = \frac{1au}{d}$$

1 parsec is defined as 1 arcsecond (1/3600 of degree) for parallax

- Using the Gaia telescope: high-precision up to 1kpc; up to 10kpc but unreliable
- Usually more precise for closer objects due to larger angle
- Has been used by astronomers for much longer as it is much more crude

(demonstration on board)

Distance - Brightness

$$b = \frac{L}{4\pi d^2}$$

b : [MT⁻³] , [Wm⁻²]

L : [ML²T⁻³] , [W]

d : [L] , [m]

$$\text{MT}^{-3} = \text{L}^2 / \text{ML}^2\text{T}^{-3} = \text{MT}^{-3}$$

b : apparent brightness (power per unit area from distance d from source)

L : luminosity (power of the star)

d : distance from star

We can calculate the distance from us and a light emitting source by measuring the apparent brightness and luminosity.

We can measure the apparent brightness through instruments in Earth, so we need the luminosity.

Distance - Cepheids

Cepheids are stars with cycles of brightness, and we can calculate the luminosity of them with the period measured using the Period-Luminosity relationship (Leavitt's Law).

$$M_H = -3.26(\log_{10} P - 1) - 5.93$$

P: period in days

M_H : absolute magnitude in H-band (near infrared) ; logarithmic expression of luminosity

Accurate around 0.05 Mpc ~ 40 Mpc due to clustering of stars (Hubble Space Telescope, James Webb Space Telescope)

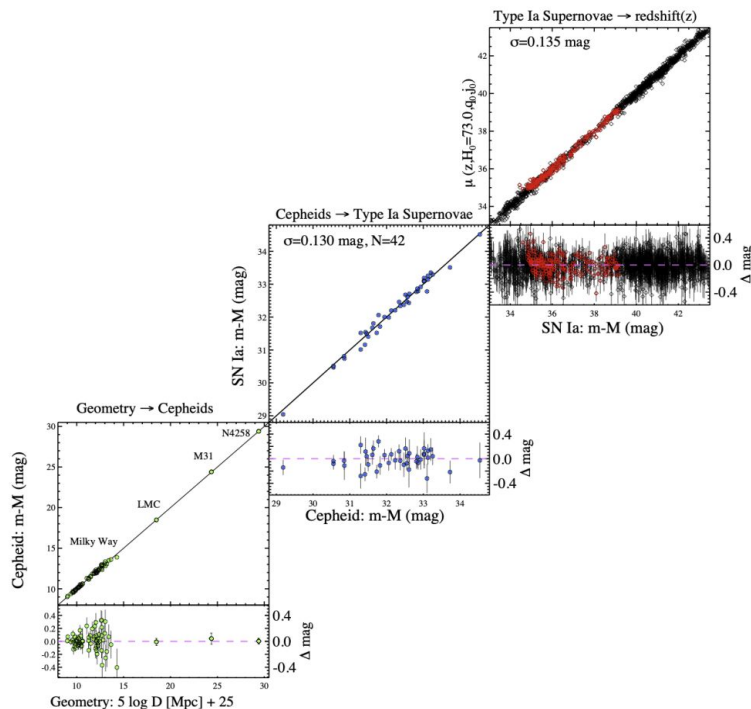
Distance - Type 1a Supernova

We are also able to calculate the luminosity of Type 1a Supernovae

- Type 1a Supernova is an explosion caused by a white dwarf reaching critical mass (Chandrasekhar mass), making every supernovae similar to each other
- There is also a relationship between peak luminosity and the time for Supernovae to dim called light curve, increasing precision
- The light curve equation shows the magnitude relationship of dimming rate and peak luminosity (Phillips relation)

40 Mpc $\sim$ 300-400 Mpc for measuring Hubble constant, possibly up to thousands of Mpc

Result



From the Cosmic Distance Ladder method,
we are able to get

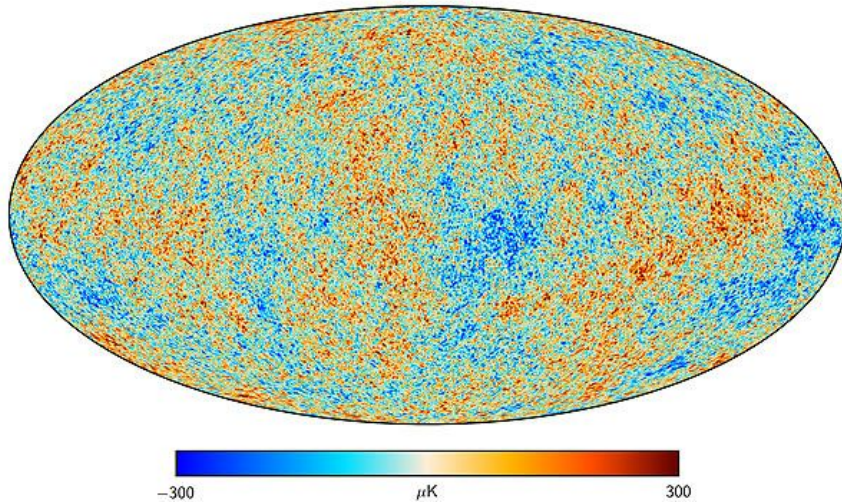
$$H_0 = 73.50 \pm 0.81 \text{ km s}^{-1} \text{ Mpc}^{-1}$$

from the recent H0DN collaboration

(The Local Distance Network: a community
consensus report on the measurement of the
Hubble constant at $\sim 1\%$ precision)

Another way to measure
Hubble Constant:
Cosmic Microwave Background (CMB)

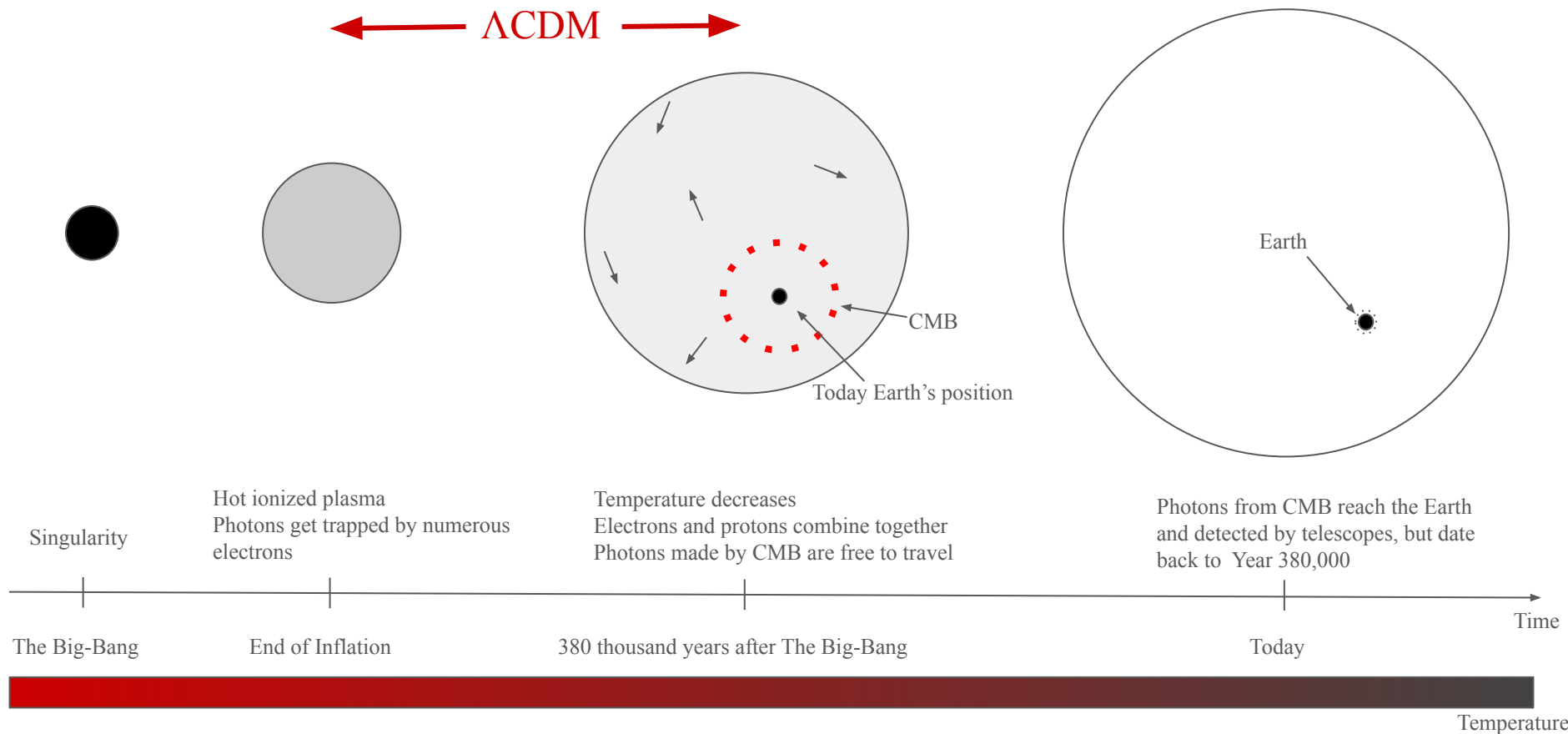
What is Cosmic Microwave Background (CMB)?



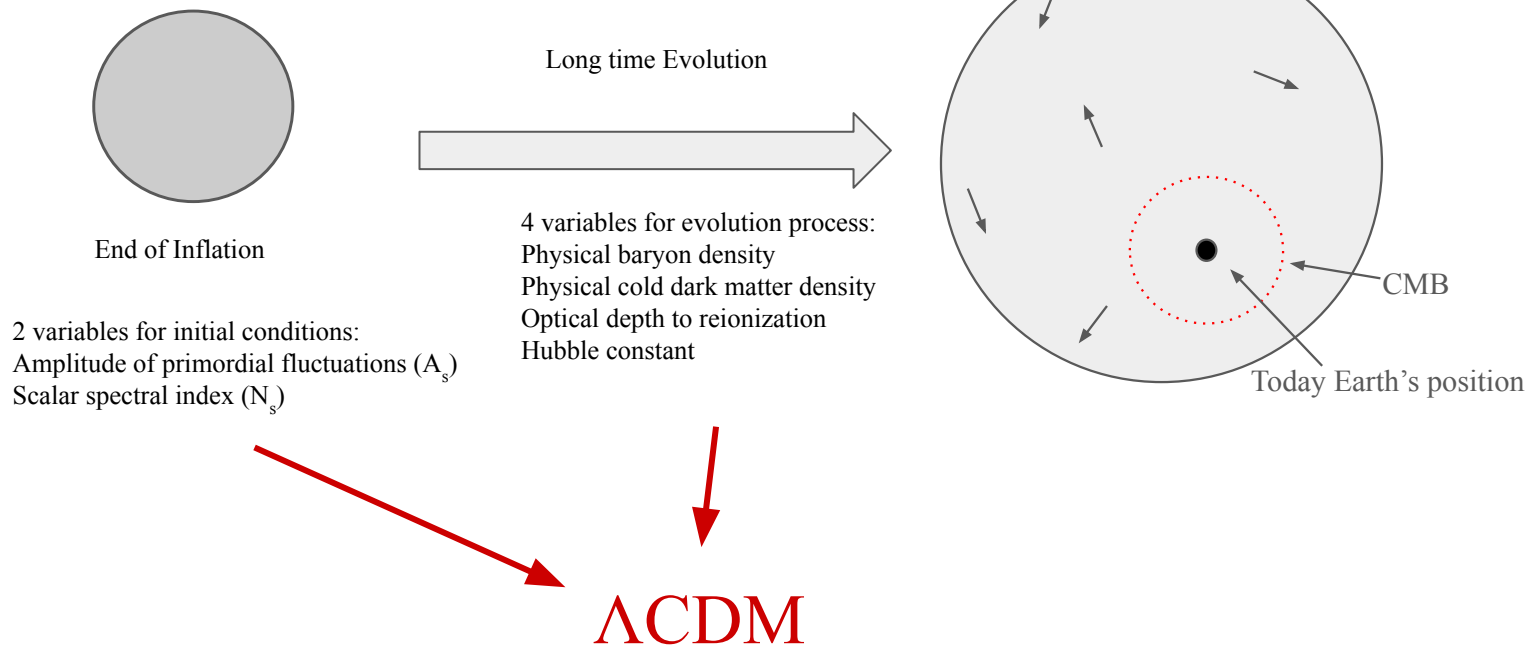
Temperature $\longleftrightarrow$ Photon's energy

- Full-sky map of CMB temperature anisotropies, showing temperature fluctuations of approximately $\pm 300 \mu\text{K}$ across the sky.
- A telescope detects microwave photons from every direction in the sky.
- Since the photon spectrum is almost a perfect blackbody, the measured photon energy can be expressed as an equivalent temperature.

How CMB forms?



Λ CDM(Cold Dark Matter) Model



Assumptions:

The universe is relatively uniform over large areas

The theory of relativity is true

We only have baryons, dark matter, dark energy and radiation

Quantum fluctuations seed the growth of large scale structure

How does CMB help us to find the Hubble Constant?

The CMB allows us to measure the Hubble Constant because in the early universe, temperature is directly related to the density of matter. The density of the Universe can therefore be expressed in this equation using Friedmann's equation, showing that the density of the universe is inversely proportional to the Hubble Constant.

$$\Omega := \frac{\rho}{\rho_c} = \frac{8\pi G\rho}{3H^2}.$$
$$\frac{ML^{-3}}{ML^{-3}} = \frac{C \cdot M^{-1}L^3T^{-2} \cdot ML^{-3}}{T^{-2}} = 0$$

Friedmann's equation showing the relationship between the Hubble constant and the density of the universe.

Dimensional analysis of the Friedmann equation showing that the density parameter is a dimensional analysis.

Numerical Estimation



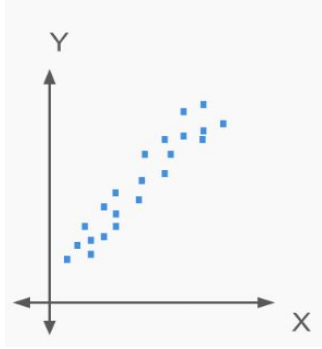
Scientists made their first observation of CMB in 1964, and latest to almost today. They are definitely not the same photons. Then how do they get the same result?

Lets compare the time:

$$\frac{\Delta t_{\text{human observation span}}}{T_{\text{total flight time of CMB photons}}} = \frac{2026 - 1964}{1.3787 \times 10^{10} \text{ years}} = \frac{62}{1.3787 \times 10^{10}} \approx 4.5 \times 10^{-9}$$

The ratio is close to 0. They are basically the same!

Logic Analogy



Scatterplot

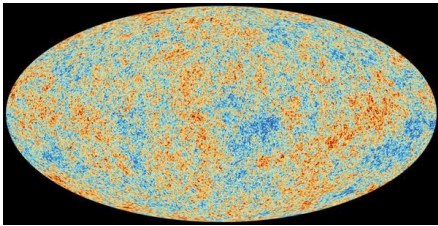
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Linear Function Model

$$y = kx + b$$



k, b



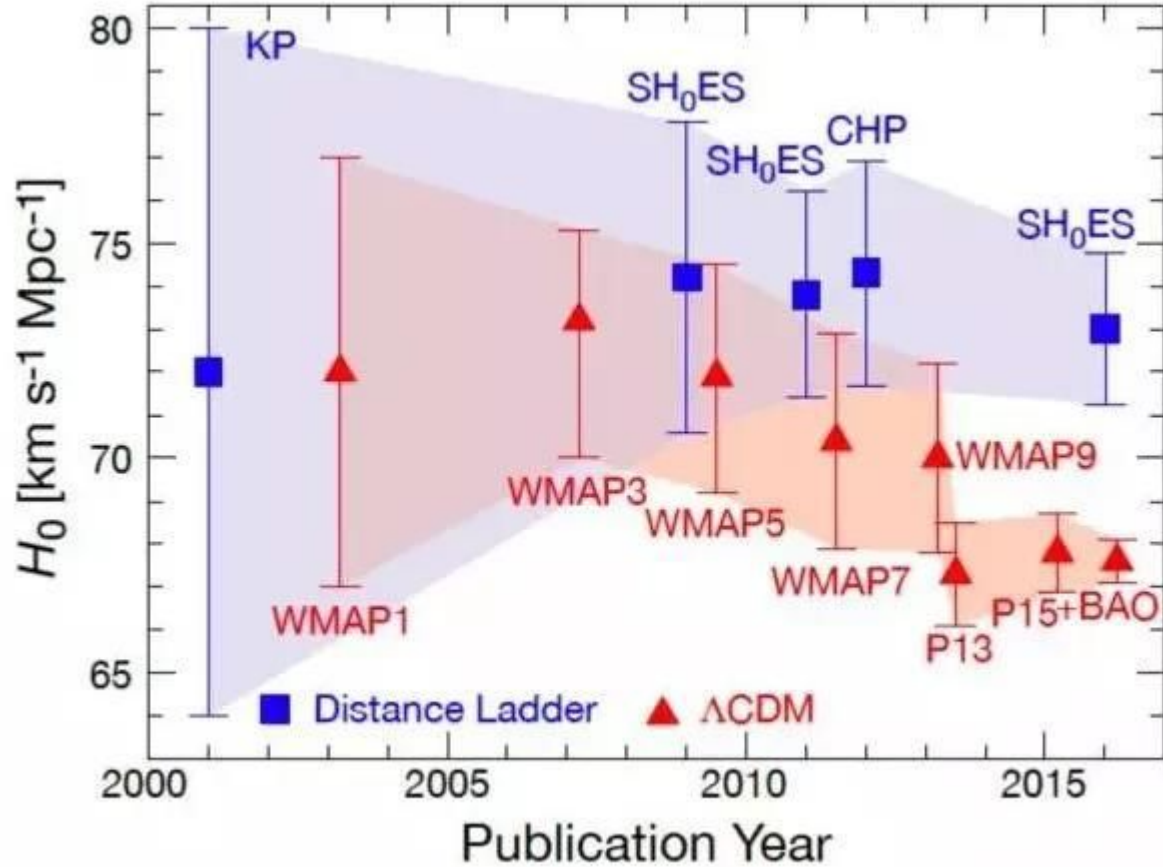
CMB

+

Λ CDM



$H_0 = 67.6 \text{ km/s/Mpc}$



What we don't know: Possible Reasons for the Discrepancy

1. Distance ladder is wrong-systematic error/luminosity depends on metallic composition/... (This is because certain metals present change the apparent luminosity of certain structures.)
2. CMB is wrong- Λ CDM model is incomplete and we need new physics
3. We live in a local cosmic void – the local Universe is not representative of the average Universe, so local observations give a larger H_0

Thank You

Derivation for Cosmic Redshift (only if asked)

$$a(t_{emit}) = a(t_{now} - \Delta t) \approx a(t_{now}) - a'(t_{now})\Delta t = 1 - H_0 \Delta t$$

$$1 + z = \frac{\lambda_{obs}}{\lambda_{emit}} = \frac{a_{now}}{a_{emit}} = \frac{1}{a_{emit}}$$

$$1 + z = \frac{1}{1 - H_0 \Delta t} \approx 1 + H_0 \Delta t$$

$$z \approx H_0 \Delta t = H_0 * \frac{d}{c}$$

$$cz \approx H_0 d = v$$